Wipro Garnet
If light bulbs are not your thing, check out these really cool dimmable LED tuble lights from Wipro. **https://dgit.in/may20-03**

Amazon Solimo
These Amazon Solimo light bulbs also use the ESP8366 board, and this DIY works on them too. **https://dgit.in/may20-04**

MQTT messages, but HASS provides a nice interface of control.

REQUIREMENTS
For flashing, you will need an up to date Debian-based Linux computer with Wi-Fi (For example, an old laptop, or a Raspberry Pi, or even a live disk. A VM will probably not work.)

- For the HASS server, an up to date Linux computer with Wi-Fi (An old laptop, Raspberry Pi, or even a VM. Note that a live environment is not optimal as you will have to install the server at every reboot.)
- A Wi-Fi device (A phone, a tablet, or another laptop connected to Wi-Fi)
- Docker / alternate method to install HASS (Optional, only if you want app UI and automations)
- Tuya-convert
- A compatible Wi-Fi bulb
- A few hours on a good weekend!
- Home Assistant: Home-Assistant is a free and open source home automation and control solution. There are many ways and compatible devices to install it. Personally, we ran it in a python virtual environment on our home server. But for this guide, we recommend the docker method, as this is just a single command and requires minimal configuration.

TUYA AND TUYA-CONVERT
Tuya is a China-based manufacturer of smart electronics. As long as you have a Tuya device, you can flash custom firmware for the device in an OTA style update. No soldering required! Some Tuya devices available on the market include the Wipro Garnet line of smart lights, and the Amazon Solimo brand of smart LEDs.

WARNING: Remember to check your exact model number against the device list on Tasmota wiki before purchase. From personal experience, we can only confirm that Wipro Garnet NS9001 works.

The first time a Tuya device connects to the internet, it calls home to fetch updates. We use a program called tuya-convert which pretends to be the update server for the bulb, and installs a firmware of our choice into the device. We need a Debian-based computer because the program runs apt-get in the background. This can be modified for use with other distributions of Linux, but this is not covered in this article. While this article is written with a Debian base in mind, it can easily be modified for other Linux distributions.

TASMOTA
ESP8266 is a Wi-Fi enabled microcontroller commonly used in IoT devices. Tasmota is a free and open source custom firmware for ESP8266 based devices. It works for a wide range of devices including bulbs, switches, and kettles. You can configure the firmware to work for your device by specifying a device template, so it knows what device it is installed on. With Tasmota installed, you get features such as full brightness and colour control, timed executions, etc, with complete control over your device. No proprietary spying blackboxes. Sadly, you lose the ability to voice control and app control the bulb with just Tasmota, as it provides only a web interface and an MQTT interface, but the functionality can be added back with Home-Assistant.

The App control and Voice control functionality is provided by Home Assistant, everything else by Tasmota.

UPDATES
You should first update your computer

```
sudo apt-get update && sudo apt-get upgrade
```

If you are running a live environment, you may need to enable universe repository before running updates.

```
sudo add-apt-repository universe
```

MQTT
Next, install an MQTT broker so that the smart bulb and HASS can talk to each other. (Do this on the 'server' PC.)

```
sudo apt-get install mosquitto mosquitto-clients
```

Now, we need to open the MQTT port on your computer firewall.

```
sudo ufw allow 1883; sudo ufw enable
```

HOME ASSISTANT
We need to remove any conflicting packages on your system (server PC).

```
sudo apt-get remove docker docker-engine docker.io containerd runc
```

Then, we will need to install the packages required for docker

```
sudo apt-get install -y software-properties-common apparmor-utils apt-transport-https avahi-daemon ca-certificates curl dbus jq network-manager socat ufw
```

Modem manager service needs to be disabled since it can cause conflicts

```
sudo systemctl disable ModemManager.service
```

Install Docker Community Edition-

```
curl -fsSL get.docker.com | sh
```